

Preliminaries

In this chapter we review the basic language of algebraic topology and differential geometry that is needed to discuss Seiberg-Witten theory and the Donaldson theorem. In the differential realm, we assume familiarity with smooth manifolds, de Rham cohomology, vector bundles, and some Riemannian geometry. In algebraic topology, we assume familiarity with singular and cellular homology.

Gauge theory Gauge theory is the heart of Seiberg-Witten theory. The name comes from physics, where gauge fields are fields with symmetries and redundant degrees of freedom. Mathematically, the study of gauge fields is done in terms of principal bundles, connections, curvature, and associated vector bundles.

Principal bundles, connections and curvature The basic objects of gauge theory are principal bundles.

definition[Principal G -bundle] Let M be a smooth manifold and let G be a Lie group. A principal G -bundle over M is a smooth manifold P together with a smooth surjective map $\pi: P \rightarrow M$ and a right action of G on P satisfying equivariance and local triviality conditions.